

RubricEye Exam

Answer Booklet

Name: _____

Candidate ID: _____

Date: _____ Exam Version: RE-01

Answer only the questions you choose.

Keep all writing and drawings inside the bordered areas.
For diagram/flowchart responses, use clear labels and arrows.

Q2(i)

4 marks

Newtons second law of motion; states that acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. The acceleration occurs in the direction of applied force.

Mathematical expression

$$F = ma$$

Where

F = Force (N, Newton)

m = Mass (Kg)

a = acceleration (m/s^2)

Q2(ii)

4 marks

1- RAM stands for Random Access Memory
 2- It is temporary / Volatile memory
 3- Data is lost when computer is off
 4- It stores data & program under use (currently).

1- ROM stands for Read only memory
 2- It is permanent / non-volatile memory
 3- Data remained store in computer
 4- It store important instruction require to start / operate the computer.

Example; DRAM, SRAM

Example; PROM, EPROM

Q2(iii)

4 marks

The Nucleus is control centre of eukaryotic cell

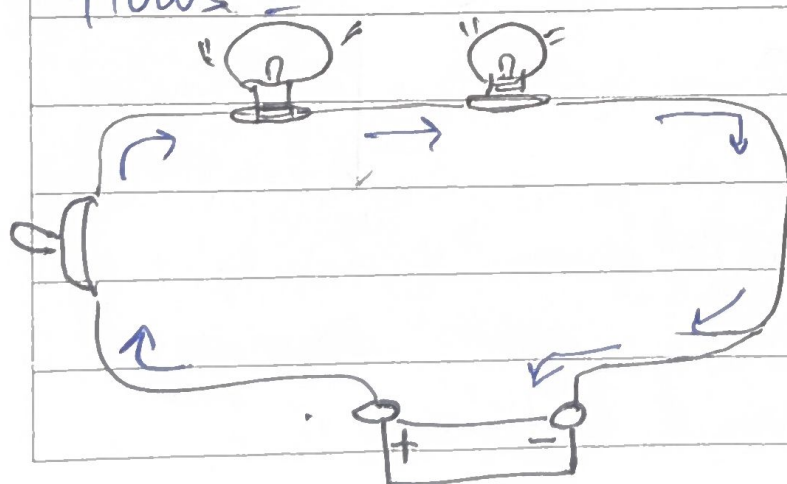
Its Main functions are

- i- It has cell's genetic material/DNA
- ii- It also regulates protein synthesis through genetic information.
- iii- It controls cell's activities such as, growth, metabolism & reproduction.
- iv- It helps in cell division by passing genetic material to daughter cells.

Q2(iv)

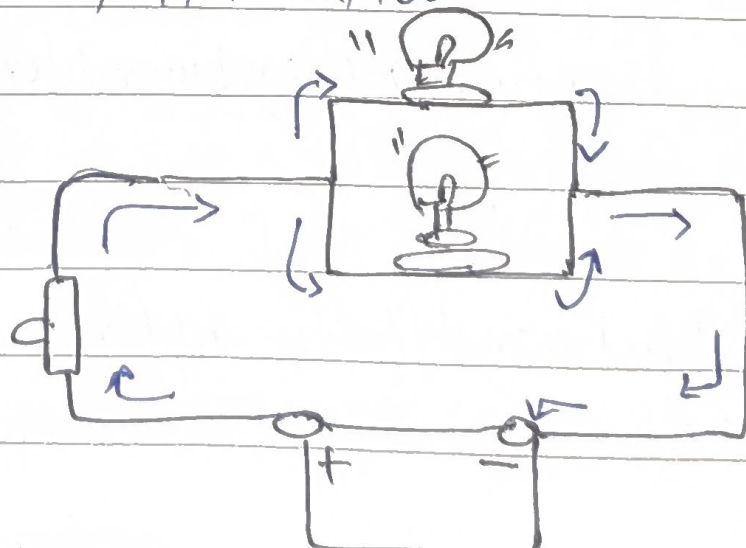
4 marks

Series Circuit in this electrical components are connected one after another in a single path. The same current flows.



parallel circuit in which components are connected in separate branches.

The current has more than one path flow.



4 marks

Q2(v)

INPUT DEVICES

1- KEYBOARD

It is used to enter text, numbers and commands into the computer

2- Mouse It is used to point, select click $\rightarrow$ control object on the screen

OUTPUT DEVICES

1- Monitor It display text, images

video $\rightarrow$ other information from computer.

2- ~~Mouse~~ Printer: It produce the large hard copy of digital documents

Q2(vi)

4 marks

Four forces acting on the box

1- Upward force

The surface exerts upward normal reaction on box

2- Downward force

$\rightarrow$ Weight / gravitational force

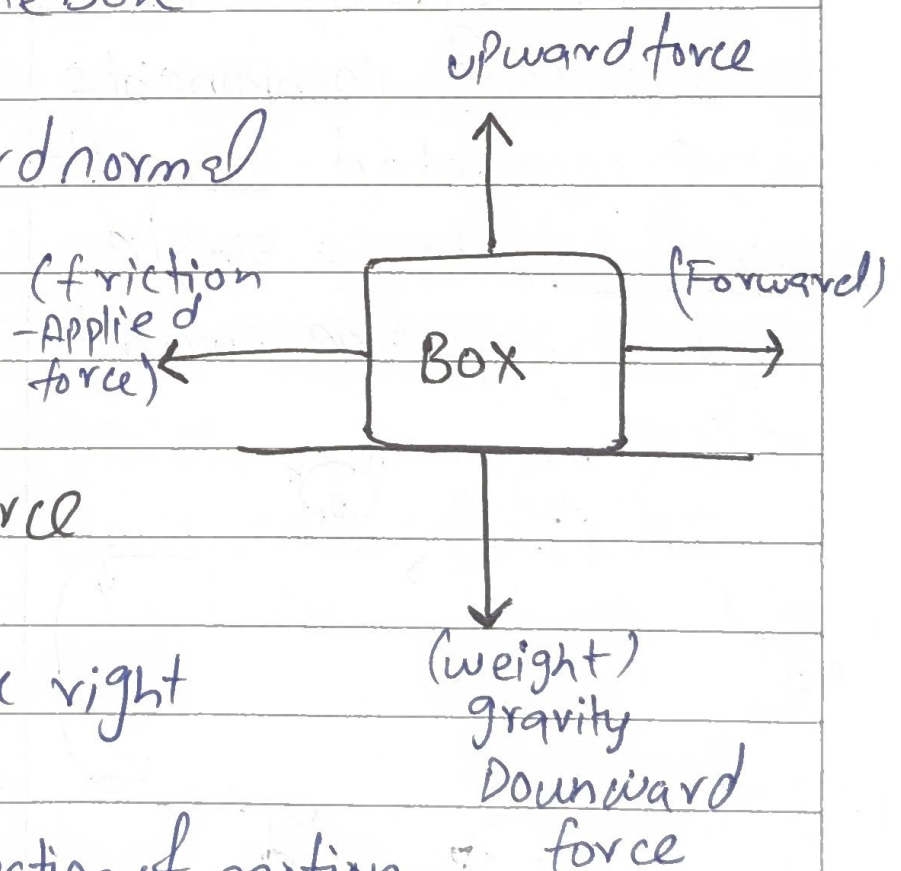
3- Forward / Rightward force

$\rightarrow$ Applied force -

Force that pull the box right

4- Backward / Leftward force

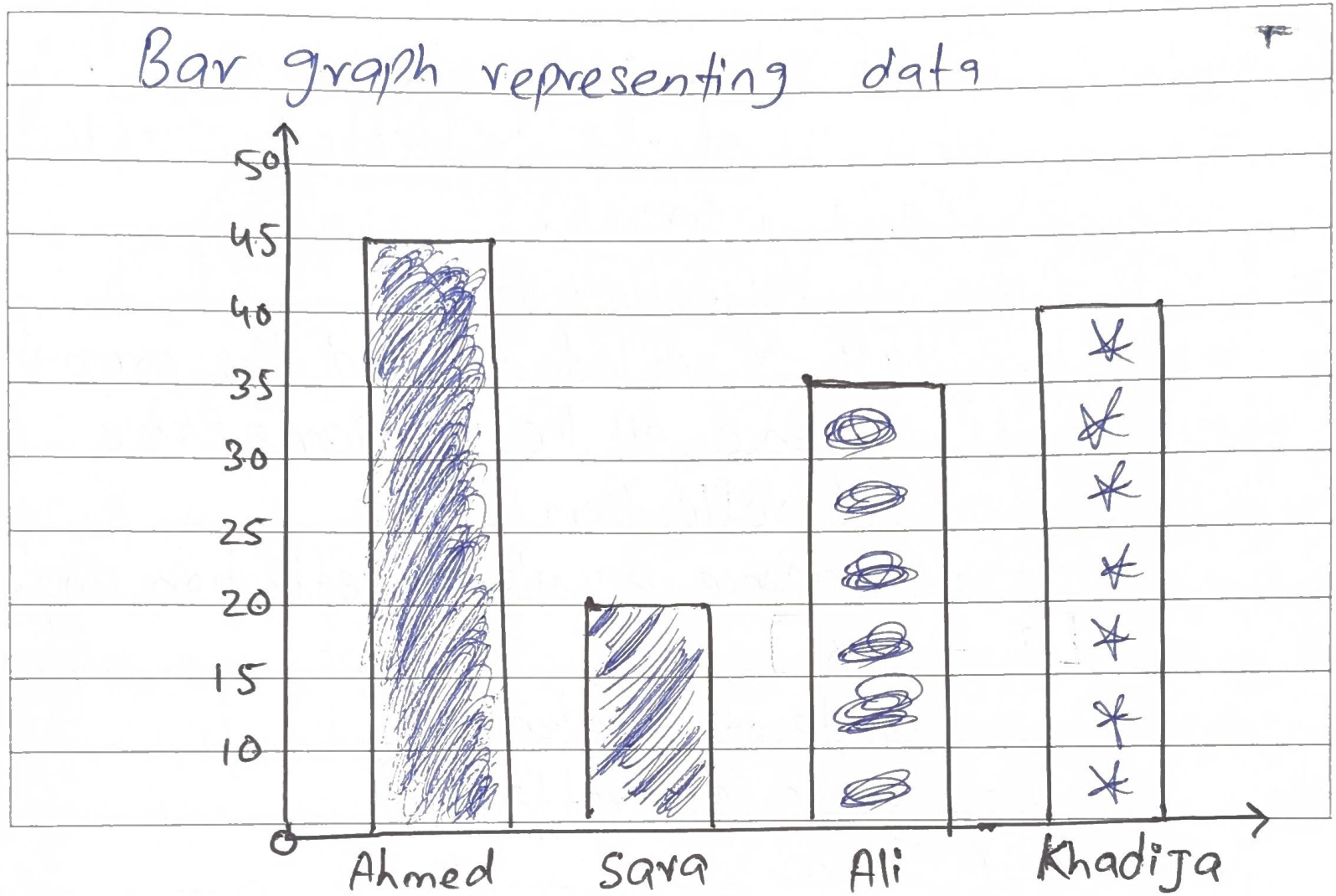
frictional opposite to direction of motion



RubricEye Exam

Q2(vii)

4 marks



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Q3(A)

The law of reflection; describe how light behaves when it strikes a reflected surface such as plane mirrors.

First Law of reflection &~

The incident ray, reflected ray, and the normal at point of incidence all lie in same plane

Second law of reflection

The angle of incidence equals to reflection angle
 $[i = r]$

where, i = angle of incidence

r = angle of reflection

Important Term

Incident ray

The ray of light that strikes the mirror.

Reflected ray

The ray of light that bounces back from the mirror

Normal

The imaginary line drawn perpendicular (90°) to the mirror at point of incidence

Angle of Incidence

The angle between incident ray & the normal

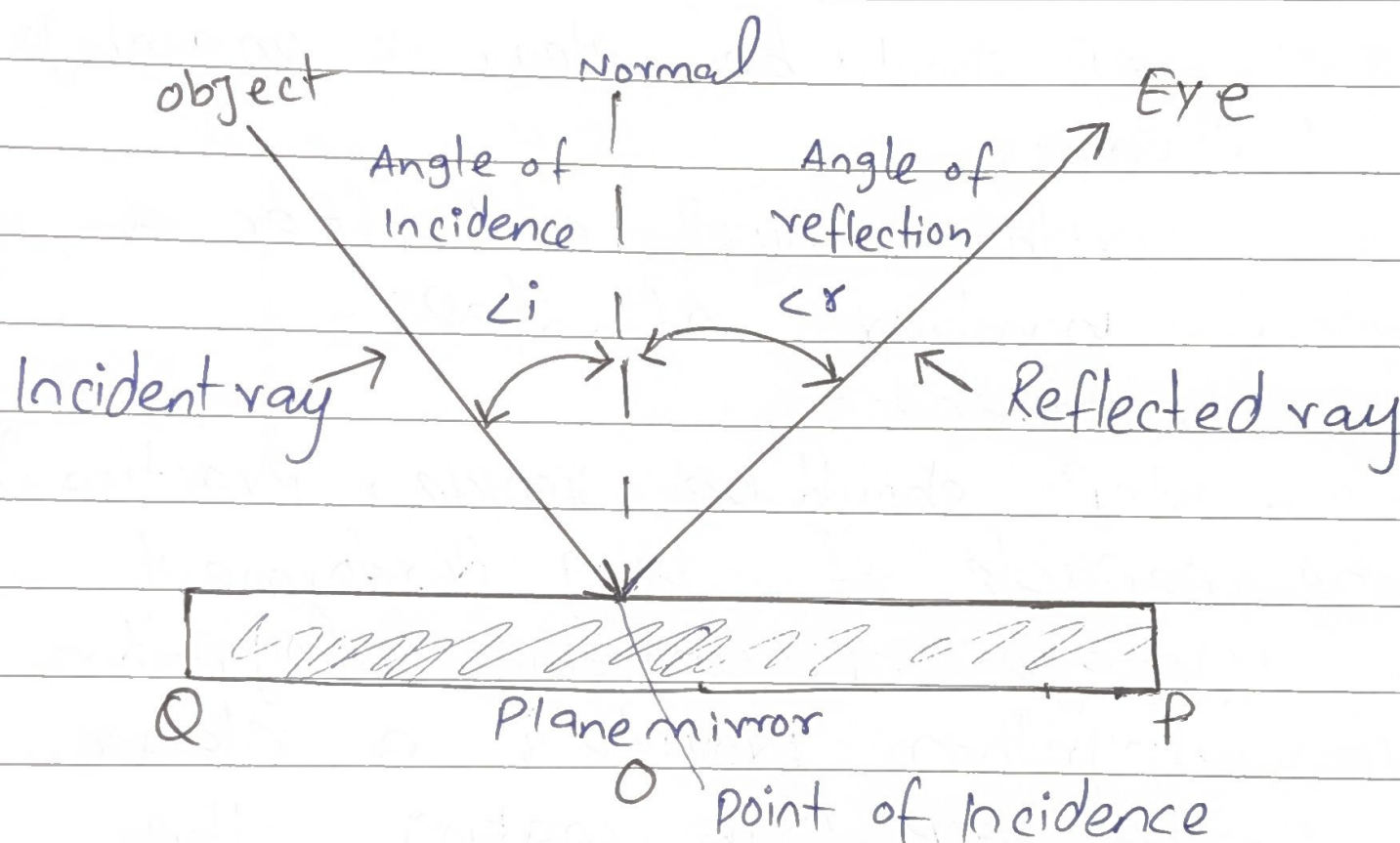
Angle of reflection

The angle between reflected ray and normal

Q3(B)

7 marks

Following diagram represent the (law of reflection) reflection from a plane mirror -
According to law of reflection:



Horizontal line = Plane mirror

vertical line = Normal

lower slanting ray = Incident ray

upper slanting ray = reflected ray

i = Angle of Incidence

r = Angle of reflection

Q4(A)

8 marks

An algorithm is a step-by-step set of instructions used to solve a problem or perform a task

Three characteristics of good algorithm:

1 - Definiteness

Every step should be clear & unambiguous

2 - Finiteness

The algorithm must end after a finite number of steps -

3 - Effectiveness

Each step should be simple, practical and capable of being performed

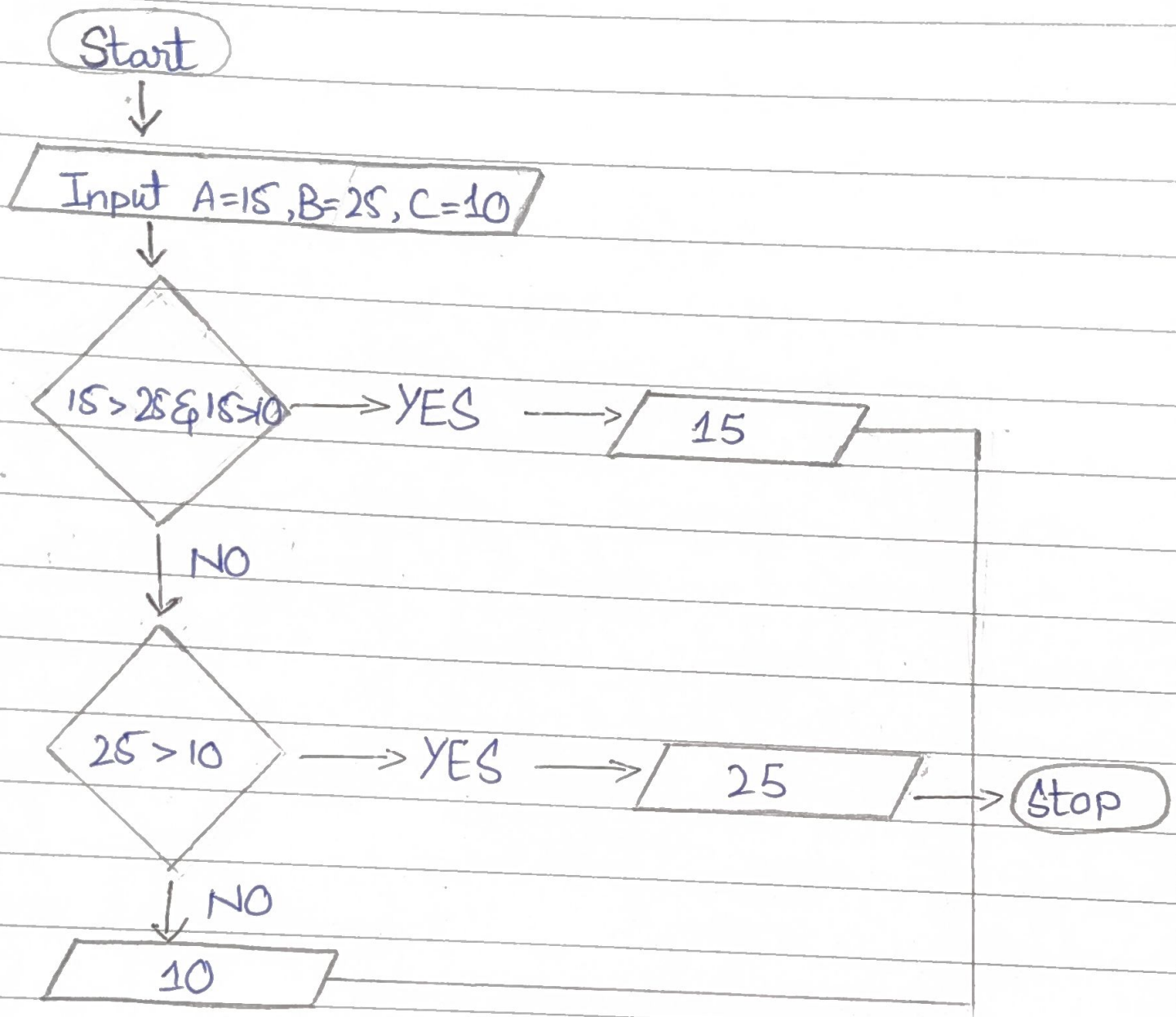
One Advantage of writing algorithms -

An algorithm provides a clear step-by-step plan, making the problem easier to understand

the helping the programmer to develop the correct program with fewer errors -

Q4(B)

7 marks



ROUGH WORK

Handwritten notes and calculations on lined paper, including:

- Top right: 1.2
- Center: $0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0$
- Below center: $1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0$
- Bottom right: $1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0$